SMART INDIA HACKATHON 2025

- **Problem Statement ID –** 25045
- **Problem Statement Title-**Blockchain-Based Supply Chain Transparency for Agricultural Produce
- **Theme-** Agriculture
- **PS Category-** Software
- **Team ID-**
- **Team Name –** Code Panda

CODE PANDA

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AgriChain ZK: An Anti-Exploitation & Rural Empowerment Protocol

THE KATNI-CHHATNI CRISIS (LEGACY SYSTEM)

Smallholder Farmer

↓

Manual **Mandi** Assessment (Bias/Fraud)

↓

Arbitrary 5-10kg/Quintal Deduction

↓

Delayed Bureaucratic Payment

TRUSTLESS LEDGER INTERVENTION

Smallholder Farmer

↓

AI Grain Analyzer + IoT Oracles

↓

Immutable Smart Contract Validation

↓

Instant DBT Escrow Payout

ZK-SNARKs: Ensures MSP compliance without leaking sensitive enterprise trade routes.

AI-Hardware Oracles: Eradicates Garbage In, Garbage Out manual entry flaws.

Offline-First Sync: Bridges the **rural digital divide** via local payload caching.

TECHNICAL APPROACH

Physical-to-Digital Bridge Pipeline

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[Offline PWA]

The Farmer Node

Logs harvest data offline at the field.

Stack

React Native, SQLite / WatermelonDB, Background Sync Manager.

ledger tracing

[AI Oracle]

The Mandi Oracle

Bypasses manual data entry. Smartphone camera objectively assesses paddy moisture. ESP32 IoT sensors track transit health.

Stack

AI Visual Analyzer API, NodeMCU Transit Oracles.

ledger tracing

[Polygon EVM] [ZK-Privacy]

The Immutable Ledger

Executes automated escrow and secures data without revealing sensitive trade volumes to competitors.

Stack

Polygon Amoy Testnet, Solidity Smart Contracts, IPFS, CIRCOM Zero-Knowledge Proofs.

ledger tracing

[Verification]

The Consumer & Govt Node

Instant farm-to-fork auditing via timeline UI.

Stack

Next.js Web Dashboard, QR Cryptographic Verification.

FEASIBILITY AND VIABILITY

Bypassing Rural Constraints

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Risk/Mitigation Diagnostic Table

Rural Hurdle	Why Standard Tech Fails	Our Strategic Mitigation
Intermittent / Zero 4G Connectivity	Web3 wallets fail to sign transactions offline; data is lost.	**Offline-First Caching.** Cryptographic payloads are stored locally via WatermelonDB and autonomously executed when a signal is acquired.
The Garbage In, Garbage Out (GIGO) Flaw	Corrupt officials can just type fake poor quality metrics into the blockchain.	**AI/IoT Hardware Oracles.** Smart contracts strictly reject manual quality inputs, exclusively accepting signed data from AI grain scanners and IoT sensors.
High Compute / Enterprise Costs	Rural mandis cannot afford AWS servers or Hyperledger node maintenance.	**Storage Light Nodes (SLNs).** Polygon testnets provide zero-cost transactions, while $50 Raspberry Pi nodes use Bloom filters to validate consensus locally.

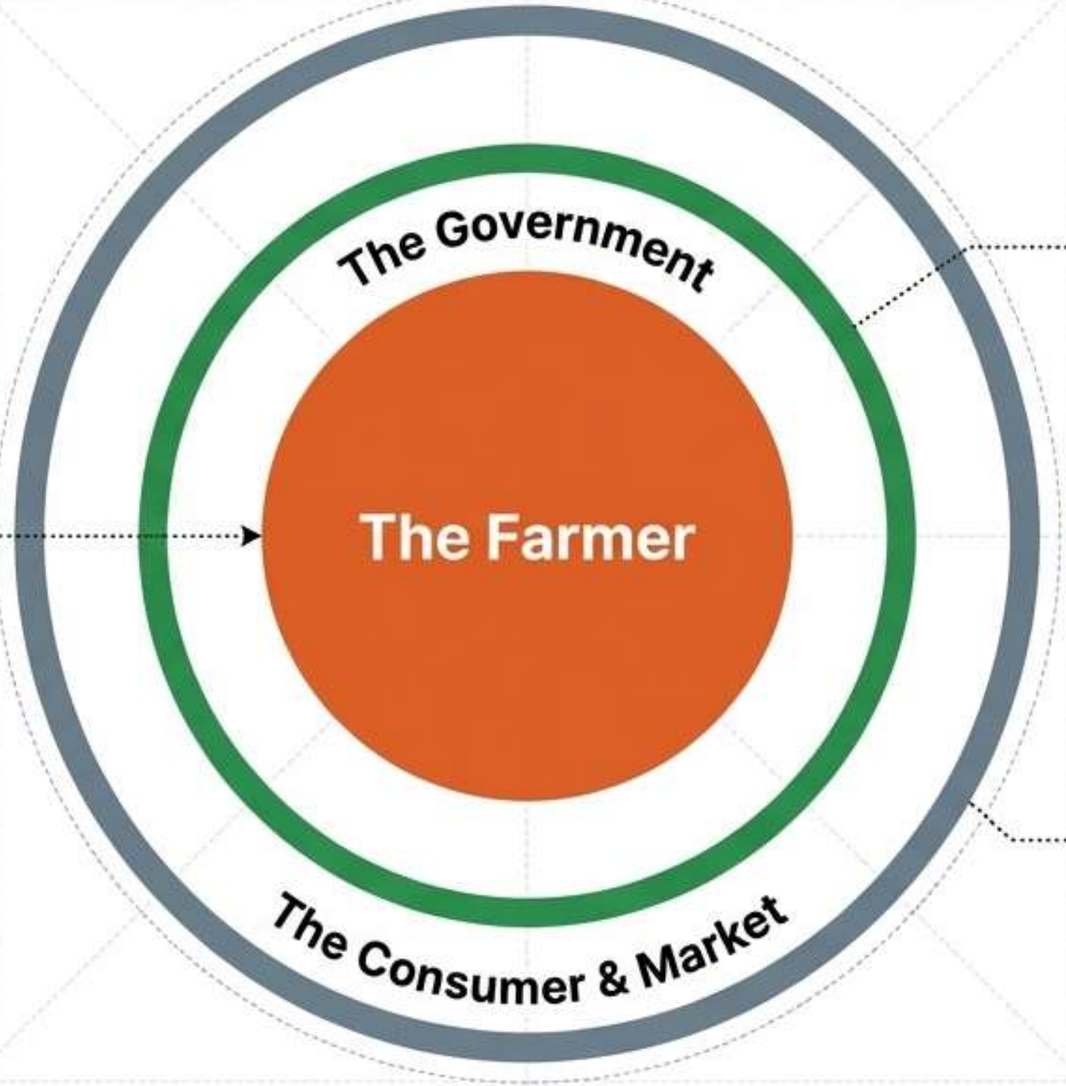
IMPACT AND BENEFITS

Multi-Dimensional Socioeconomic Value

Smallholder Farmers (Primary Beneficiaries)

Impact: Structural democratization of agricultural data.

Benefit: Permanent eradication of Katni-Chhatni extortion. Guaranteed fair price realization and unblocked working capital via instant Smart Contract payouts.



State Government & Regulators

Impact: Aligns perfectly with Odisha AI Policy 2025.

Benefit: Leak-proof administration of Minimum Support Price (MSP) funds. Total elimination of ghost beneficiaries and automated, tamper-proof procurement audits.

Consumers, Enterprises & Environment

Impact: Verification of Organic and Pesticide-Free claims.

Benefit: FDA FSMA 204 compliance for agri-enterprises. Identifies logistical bottlenecks, targeting a 20-30% reduction in post-harvest agricultural waste.

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RESEARCH AND REFERENCES

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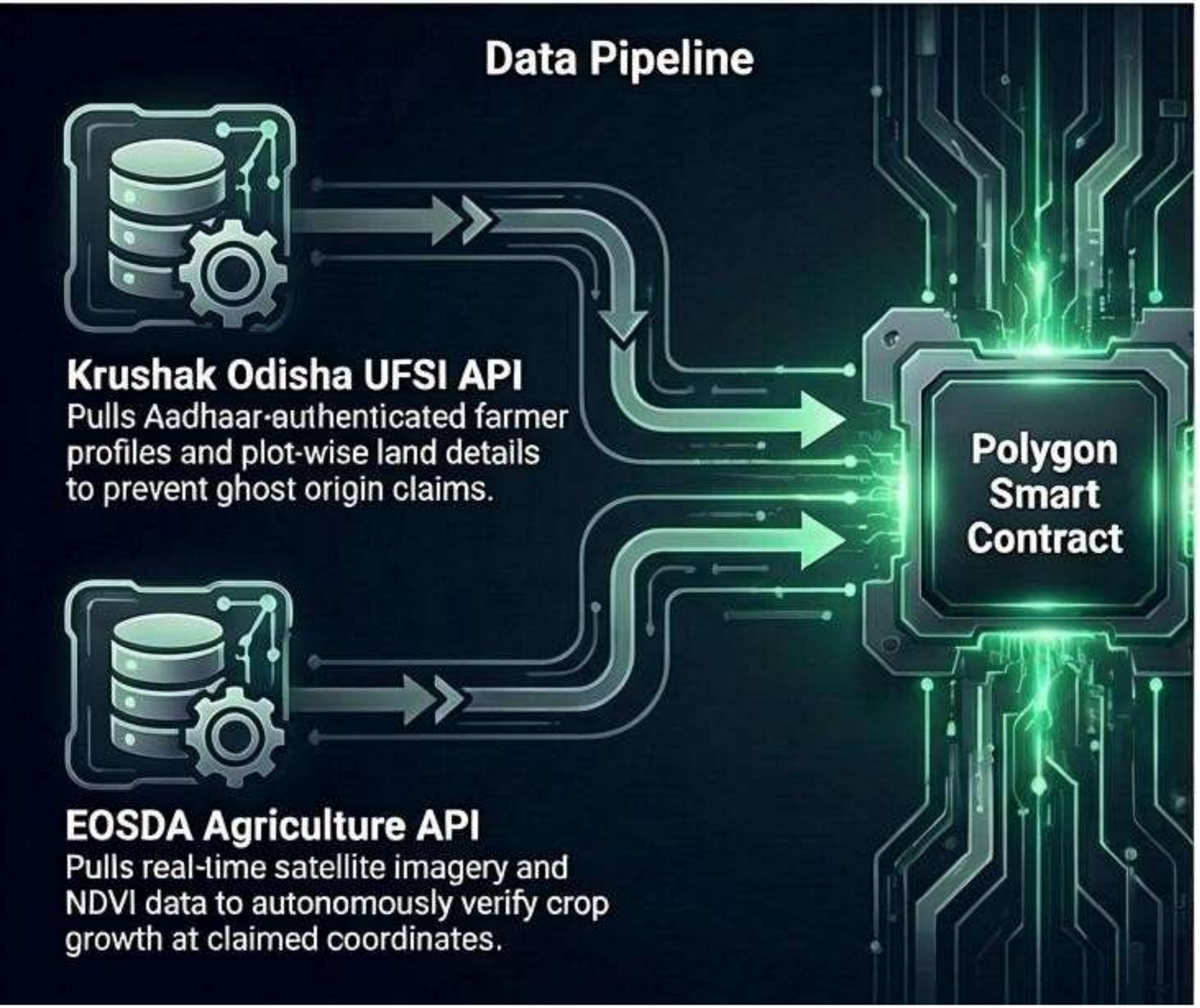

STRATEGIC ALIGNMENT & RESEARCH

Policy Alignment: Integrates guidelines from the Odisha State Data Centre Policy 2022 and Odisha AI Policy 2025.

Foundational Research: Incorporates academic models of Storage Light Nodes (SLNs) on Raspberry Pi and ZK-SNARKs for enterprise privacy.

Ground Reality: Addresses verified reports of Western Odisha farmer protests regarding miller exploitation and delayed P-PAS procurement.